SAINTGITS COLLEGE OF ENGINEERING (AUTONOMOUS), KOTTAYAM

First Series Examinations, August 2025
24BSE2113-D Partial Differential Equations, Transforms and Optimization Methods

Duration: 90 minutes | Maximum marks: 50

Part A
(Answer all questions)

1. Derive the partial differential equation from the relation $z = f(x + ay) + g(x - ay)$. (5 marks)
2. (a) Form a PDE by elimination of the arbitrary constants a and b from $(x-a)^2 + (y-b)^2 = z^2 \cot^2 \alpha$. (3 marks)
   (b) Solve using Lagrange method, $\frac{y^2 z}{x} p + xzq = y^2$. (7 marks)
3. Using method of separation of variable, solve $3u_x + 2u_y = 0$ where $u(x, 0) = 4e^{-x}$. (10 marks)

Part B
Answer all questions.

4. State all basic assumptions in the solution of one dimensional heat equation of second order and write all possible solutions under these assumptions. (5 marks)
5. A tightly stretched string of length l with fixed ends is initially in equilibrium position. Motion is governed by $\dfrac{\partial^2 u}{\partial t^2} = c^2 \dfrac{\partial^2 u}{\partial x^2}$ and is started by displacing the string into the form of the curve $y = a \sin\left(\frac{\pi x}{l}\right)$, $0 < x < l$, from which it is released at time $t = 0$. Find the displacement of any point at a distance x from one end at time t. (10 marks)
6. Derive the one dimensional heat equation using the method of separation of variables. Also pick the acceptable solution among the possible solutions with proper justification. (10 marks)